
Supplementary Materials for: On Calibration of Modern Neural Networks

S1. Further Information on Calibration Metrics

We can connect the ECE metric with our exact miscalibration definition, which is restated here:

$$\mathbb{E}_{\hat{P}} \left[\left| \mathbb{P} \left(\hat{Y} = Y \mid \hat{P} = p \right) - p \right| \right]$$

Let $F_{\hat{P}}(p)$ be the cumulative distribution function of $\hat{P}$ so that $F_{\hat{P}}(b) - F_{\hat{P}}(a) = \mathbb{P}(\hat{P} \in [a, b])$. Using the Riemann-Stieltjes integral we have

$$\begin{aligned} & \mathbb{E}_{\hat{P}} \left[\left| \mathbb{P} \left(\hat{Y} = Y \mid \hat{P} = p \right) - p \right| \right] \\ &= \int_0^1 \left| \mathbb{P} \left(\hat{Y} = Y \mid \hat{P} = p \right) - p \right| dF_{\hat{P}}(p) \\ &\approx \sum_{m=1}^M \left| \mathbb{P}(\hat{Y} = Y \mid \hat{P} = p_m) - p_m \right| \mathbb{P}(\hat{P} \in I_m) \end{aligned}$$

where I_m represents the interval of bin B_m . $\left| \mathbb{P}(\hat{Y} = Y \mid \hat{P} = p_m) - p_m \right|$ is closely approximated by $|\text{acc}(B_m) - \hat{p}(B_m)|$ for n large. Hence ECE using M bins converges to the M -term Riemann-Stieltjes sum of $\mathbb{E}_{\hat{P}} \left[\left| \mathbb{P} \left(\hat{Y} = Y \mid \hat{P} = p \right) - p \right| \right]$.

S2. Further Information on Temperature Scaling

Here we derive the temperature scaling model using the entropy maximization principle with an appropriate balanced equation.

Claim 1. *Given n samples' logit vectors $\mathbf{z}_1, \dots, \mathbf{z}_n$ and class labels $y_1, \dots, y_n$, temperature scaling is the unique solution q to the following entropy maximization problem:*

$$\begin{aligned} & \max_q \quad - \sum_{i=1}^n \sum_{k=1}^K q(\mathbf{z}_i)^{(k)} \log q(\mathbf{z}_i)^{(k)} \\ & \text{subject to} \quad q(\mathbf{z}_i)^{(k)} \geq 0 \quad \forall i, k \\ & \quad \sum_{k=1}^K q(\mathbf{z}_i)^{(k)} = 1 \quad \forall i \\ & \quad \sum_{i=1}^n z_i^{(y_i)} = \sum_{i=1}^n \sum_{k=1}^K z_i^{(k)} q(\mathbf{z}_i)^{(k)}. \end{aligned}$$

The first two constraint ensure that q is a probability distribution, while the last constraint limits the scope of distributions. Intuitively, the constraint specifies that the average true class logit is equal to the average weighted logit.

Proof. We solve this constrained optimization problem using the Lagrangian. We first ignore the constraint $q(\mathbf{z}_i)^{(k)}$ and later show that the solution satisfies this condition. Let $\lambda, \beta_1, \dots, \beta_n \in \mathbb{R}$ be the Lagrangian multipliers and define

$$\begin{aligned} L = & - \sum_{i=1}^n \sum_{k=1}^K q(\mathbf{z}_i)^{(k)} \log q(\mathbf{z}_i)^{(k)} \\ & + \lambda \sum_{i=1}^n \left[\sum_{k=1}^K z_i^{(k)} q(\mathbf{z}_i)^{(k)} - z_i^{(y_i)} \right] \\ & + \sum_{i=1}^n \beta_i \sum_{k=1}^K (q(\mathbf{z}_i)^{(k)} - 1). \end{aligned}$$

Taking the derivative with respect to $q(\mathbf{z}_i)^{(k)}$ gives

$$\frac{\partial}{\partial q(\mathbf{z}_i)^{(k)}} L = -nK - \log q(\mathbf{z}_i)^{(k)} + \lambda z_i^{(k)} + \beta_i.$$

Setting the gradient of the Lagrangian L to 0 and rearranging gives

$$q(\mathbf{z}_i)^{(k)} = e^{\lambda z_i^{(k)} + \beta_i - nK}.$$

Since $\sum_{k=1}^K q(\mathbf{z}_i)^{(k)} = 1$ for all i , we must have

$$q(\mathbf{z}_i)^{(k)} = \frac{e^{\lambda z_i^{(k)}}}{\sum_{j=1}^K e^{\lambda z_i^{(j)}}},$$

which recovers the temperature scaling model by setting $T = \frac{1}{\lambda}$. $\square$

Figure S1 visualizes Claim 1. We see that, as training continues, the model begins to overfit with respect to NLL (red line). This results in a low-entropy softmax distribution over classes (blue line), which explains the model's overconfidence. Temperature scaling not only lowers the NLL but also raises the entropy of the distribution (green line).

S3. Additional Tables

Tables [S1](#), [S2](#), and [S3](#) display the MCE, test error, and NLL for all the experimental settings outlined in [Section 5](#).